

David Ibarra Luna

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SUMMARY

Industrial Chemist with a Master's in Materials Science, specialized in nanomaterials synthesis applied to water quality assurance. Experienced in interdisciplinary research combining materials development, contaminant analysis, and data processing with Python, aimed at solving real-world environmental challenges.

PROFESSIONAL EXPERIENCE

PhD Student / Research Staff - ICMAB, CSIC (Barcelona, Spain) Sep 2025 – Present

– Research staff in the Nanomol group under the supervision of Marta Mas-Torrent and Núria Crivillers, working on organic materials for electronic and sensing applications.

Research stay at institutes of CSIC (Barcelona, Spain) Sep 2024 – Dec 2024

– Developed carbon nanomaterials for optoelectronic nitrate sensors in water.
– Analyzed and processed spectral data from characterization techniques (electron microscopy and spectrophotometry) using Python, accelerating the interpretation of results.

Robotics teacher (High School) Aug 2023 – Jun 2025

– Taught robotics and programming (Python) to high school students using the STEM-based FIRST Lego League methodology, fostering technical and problem-solving skills through hands-on projects.

PROJECTS

DIY Optoelectronic Sensor with ML for Water Quality Ongoing

Building a Raspberry Pi-based optoelectronic sensor for water quality analysis. Implementing ML models (PLS, SVM) in Python to optimize contaminant level detection from spectral data.

Master's Work: Data Analysis & Animated Visualization [Data Repo](#) | [Animation Repo](#)

Managed the full data process for a research article, from processing spectral data with Python (Pandas, NumPy) to creating publication-ready figures (Matplotlib). Developed a fully animated thesis defense with Manim to visually communicate complex scientific results ([view presentation here](#)).

Spectrum Data Loader - Python Package (PyPI) [Link to Project](#)

Developed and published a Python library on PyPI to streamline the loading and initial processing of spectroscopy data files.

EDUCATION

2023–2025 M.Sc. in Materials Chemistry, Autonomous University of Nuevo Leon, (Grade: 94.44/100)

2019–2022 B.Sc. in Industrial Chemistry, Autonomous University of Nuevo Leon, (Grade: 82.69/100)

PUBLICATIONS

Ibarra, David et al. (Apr. 2023). “Synthesis of Graphene Quantum Dots Coupled to Au Nanoparticles: A Facile and Versatile Route Using Different Carbon Sources”. In: *C* 9.2, p. 45. DOI: [10.3390/c9020045](#).

– (2026). “A Comparative Study of Synthesis Routes for Nitrogen-Doped Graphene Quantum Dots in Fluorescence Quenching-Based Nitrite Detection”. In: *ACS Omega* 11.27, p. 39700. DOI: [10.1021/acsomega.5c12566](#).

AWARDS

2026 ANUIES Master's Thesis Award – Water Category, for the thesis: *Development of an Optoelectronic Sensor Based on Graphene Quantum Dots for Nitrite Detection in Shallow Well Water*.

SKILLS

Computing skills Python (Pandas, NumPy, Scikit-learn, Matplotlib, ManimCE), Git, GitHub, L^AT_EX, ImageJ, MS Office Suite, Inkscape.

Scientific Skills Experimental Design, Material Characterization, Quantitative Analysis.

Languages Spanish (Native), English (Advanced), French (Beginner).